

ENVIRONMENTAL SCIENCE AND ENGINEERING**ASSIGNMENT 2****CAMPUS BIODIVERSITY EXPLORER**

Photo 5136: Original field photograph showing a flowering tree and surrounding campus vegetation.

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Department / Class	BTech CSE Cybersecurity
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Survey Location	College campus and surroundings within approximately 500 m
Date of Survey	12-08-2026

1. Introduction

Biodiversity is the variety of living organisms present in an area and the interactions among them. A college campus can function as a small urban ecosystem because it contains trees, shrubs, grasses, flowering plants, insects, birds and other animals living alongside human infrastructure.

The objective of this survey was to document naturally occurring biodiversity within approximately a 500 m radius of the campus. The supplied field photographs show a mixture of planted vegetation, lawns, tree belts, pathways and open ground together with several animals and invertebrates.

The assignment brief requires a biodiversity inventory containing common names, scientific names where possible, categories, observation locations, habitats and photographs. It also requires a survey-area map, a food chain and ecological observations. This report follows that structure and uses the supplied original photographs as photographic evidence.

Survey scope and identification approach

The photographs were treated as primary field evidence. Repeated photographs of the same organism were not counted as separate species. Scientific names are used in the photographic captions wherever the supplied photographs support the identification. Where species-level identification is not supported, the organism is reported at a broader taxonomic level rather than assigning an unsupported name.

2. Study Area

The study area is the college campus and its immediately surrounding environment within an approximately 500 m radius. The supplied photographs indicate several habitat types: paved pedestrian areas, open lawns, landscaped shrub beds, mature tree belts, ground-litter zones and flowering ornamental vegetation.

Because the supplied material does not state the exact campus coordinates or provide a surveyed campus plan, the study area is described using the observed habitat types and the required approximately 500 m survey radius.

Observed habitat types

- Tree-dominated patches providing shade, perches and nesting/roosting opportunities.
- Managed lawns and grass-covered ground providing primary producer biomass and habitat for small invertebrates.
- Shrub and hedge beds supporting flowering plants and insect activity.
- Leaf-litter and soil zones supporting ground-dwelling insects and decomposer activity.
- Paved pathways and open areas used by mobile animals such as birds, squirrels and dogs.

3. Biodiversity Inventory

The inventory consolidates distinct organisms or morphotypes supported by the supplied photographs. Repeated photographs are treated as corroborating views rather than additional species records.

No.	Common name	Scientific name	Category	Observation location
1	Domestic dog	<i>Canis lupus familiaris</i>	Mammal	Near GH-9
2	House crow	<i>Corvus splendens</i>	Bird	Rishab's Cafe
3	Domestic cattle / cow	<i>Bos taurus</i> *	Mammal	Near JP auditorium
4	Indian palm squirrel	<i>Funambulus palmarum</i>	Mammal	Between AB-1 and AB-2
5	Garden lizard	<i>Calotes versicolor</i>	Reptile	ECE block lawn
6	Spider	Unidentified spider	Arachnid	AB-1 cafe
7	Butterfly	Unidentified butterfly	Insect	AB-3
8	Red cotton-stainer type bug	<i>Dysdercus</i> sp.	Insect	SSN library

No.	Common name	Scientific name	Category	Observation location
9	Small insect	Unidentified	Insect	Ice cube cafe
10	Ixora / Jungle flame	Ixora coccinea	Flowering shrub	OAT lawn
11	Chinese hibiscus	Hibiscus rosa-sinensis	Flowering shrub	On the way to football ground
12	Spider plant	Chlorophytum comosum	Herbaceous plant	Clock Tower
13	Grass	Unidentified Poaceae	Grass	OAT lawn
14	Low hedge / groundcover shrub	Unidentified	Shrub	Near college entrance
15	Frangipani / temple tree	Plumeria rubra	Flowering tree	Near Vinayagar temple
16	Oleander	Nerium oleander	Flowering shrub	Near Law block
17	Queen's flower / Pride of India	Lagerstroemia speciosa	Flowering tree	Near AB-3
18	Broad-leaved herb	Unidentified	Herb	On the way to Chemical Block

* The cattle photograph supports domestic cattle, but the exact breed or subspecies is not established from the photograph alone.

Taxonomic diversity represented

The photographs cover multiple major groups requested in the assignment: mammals, birds, reptiles, insects, arachnids, flowering shrubs, flowering trees, herbs and grasses. This demonstrates that the campus is not simply a collection of ornamental plants; it also supports mobile fauna and small invertebrates that use vegetation and ground habitats.

4. Survey Area and Observation Context

The survey was intended to cover the campus and its surroundings within an approximately 500 m radius, as specified in the assignment brief. The supplied photographs represent observations from campus vegetation, open areas, pathways and landscaped zones. Exact landmark names and coordinates were not provided in the source material.

- A – Main entrance / open pedestrian area
- B – Canteen or seating area
- C – Hostel/residential edge
- D – Tree belt and lawn
- E – Pathway/drainage and landscaped edge

5. Food Chain

The following food chain uses the trophic groups represented in the field photographs. It is shown as a step-by-step energy pathway so that each trophic level and its ecological role are clear.

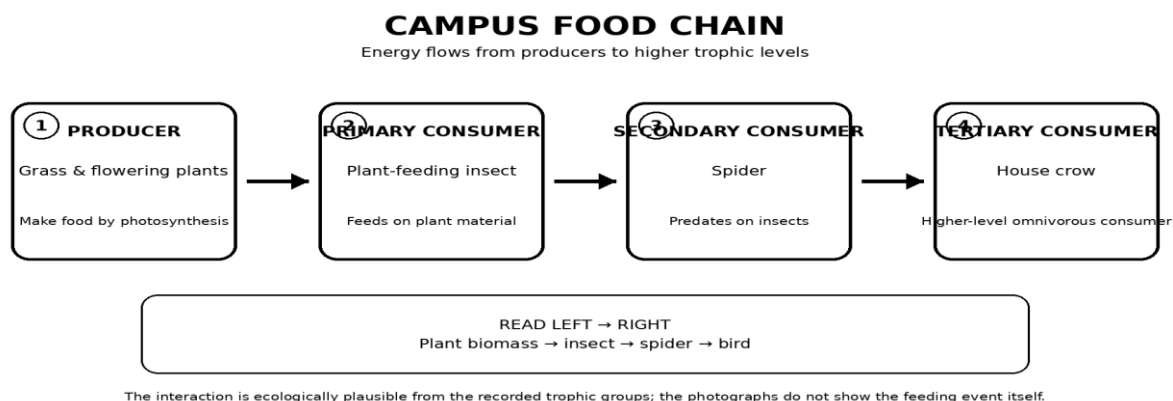


Figure 2. Representative food chain using trophic groups recorded in the survey.

Trophic level	Observed/represented organism	Ecological role
Producer	Grass / flowering plants	Photosynthesis; converts solar energy into plant biomass.
Primary consumer	Plant-feeding insect	Obtains energy from plant material or plant-derived resources.

Secondary consumer	Spider	Predates on smaller insects and transfers energy upward.
Tertiary consumer	House crow	Omnivorous bird capable of consuming a wide range of plant and animal foods.

6. Ecological Observations

Vegetation provides habitat structure: The photographs show ground-cover grasses, low shrubs, hedges and mature trees. This structural variety creates different microhabitats for insects, spiders, squirrels and birds.

Flowering plants support animal activity: Ixora, hibiscus, frangipani, oleander and other flowering plants add floral resources and increase the ecological value of landscaped areas.

Tree belts support fauna: The squirrel, garden lizard and bird observations demonstrate the importance of mature trees and shrub edges as shelter, feeding and movement areas.

Ground litter is biologically active: The photograph of red insects among dry leaves shows that leaf litter provides shelter and resources for ground-associated invertebrates.

Urban biodiversity includes generalist species: House crows and squirrels illustrate how some animals can persist close to human activity.

Managed landscapes can still have ecological value: Even maintained lawns and ornamental beds can provide food, cover and connectivity when vegetation is diverse and some soil/leaf-litter habitat remains.

Potential conservation observations

- Retain mature trees wherever safety permits, because they provide vertical habitat and shade.
- Maintain a mixture of flowering shrubs, ground cover and trees instead of a single-plant landscape.
- Avoid unnecessary disturbance of leaf litter and small invertebrate habitats.
- Reduce indiscriminate pesticide use in landscaped areas because insects and spiders are part of the campus food web.
- Maintain clean water and waste-management practices around food and open areas to reduce conflict between wildlife and people.

7. Photographic Documentation

All photographs in this section are from the supplied image archive. They are used as original field documentation; no Internet images have been inserted into the photographic plates.



Photo 4905: Canis lupus familiaris — domestic dog observed on the campus pathway/open area.



Photo 4909: Corvus splendens — house crow observed on open ground near campus seating.



Photo 5120: Bos taurus — domestic cattle observed grazing on grass.



Photo 5002: Funambulus palmarum — Indian palm squirrel at a lawn and shrub edge.



Photo 5144: Calotes versicolor — garden lizard on vegetation.



Photo 4948: Unidentified Lepidoptera — butterfly; species-level identification is not supported by the photograph.



Photo 5124: Unidentified Araneae — spider; species-level identification is not supported by the photograph.



Photo 4978: Dysdercus sp. — red cotton-stainer type bug group in leaf litter/grass.



Photo 4946: Ixora coccinea — flowering shrub (Jungle flame/Ixora).



Photo 4953: Hibiscus rosa-sinensis — Chinese hibiscus flowering shrub.



Photo 4990: Chlorophytum comosum — spider plant in a landscaped bed.



Photo 4992: Poaceae — grass ground cover; species-level identification is not supported by the photograph.



Photo 4984: Unidentified shrub — low hedge/groundcover planting; species-level identification is not supported by the photograph.



Photo 5134: *Plumeria rubra* — frangipani/temple tree flower and foliage.



Photo 5142: *Nerium oleander* — pink oleander flowers.



Photo 5132: *Lagerstroemia speciosa* — Queen's flower/Pride of India.



Photo 4925: Unidentified herbaceous plant — broad-leaved ground plant; species-level identification is not supported by the photograph.



Photo 5148: Campus vegetation habitat — mixed tree-dominated green area; no single species is represented.

8. Conclusion

The campus survey photographs demonstrate a meaningful range of biodiversity within a relatively small urban area. The evidence includes mammals, birds, reptiles, insects, spiders, grasses, herbs, shrubs and flowering trees. The diversity of vegetation creates habitat structure that supports both visible animals such as squirrels and birds and less conspicuous organisms such as insects and spiders.

The survey also shows why biodiversity conservation on campuses should not be limited to protecting large trees alone. Ground cover, leaf litter, flowering shrubs, tree belts and small patches of less-disturbed habitat all contribute to the local ecological network. Maintaining this variety can strengthen habitat availability and support food-web interactions within the campus ecosystem.

Overall, the photographic inventory provides field-based evidence of campus biodiversity and covers the required components of a biodiversity inventory, survey-area representation, food-chain construction, ecological observations and original photographic documentation.

References and Identification Sources

Assignment source

- Environmental Science and Engineering Assignment – Assignment 2: Campus Biodiversity Explorer (provided course document).

Taxonomic verification sources

- [NCBI Taxonomy – *Canis lupus familiaris* \(domestic dog\)](#)
- [NCBI Taxonomy – *Bos taurus* \(domestic cattle\)](#)

- [Indian Biodiversity Information System – House Crow \(*Corvus splendens*\)](#)
- [NCBI Taxonomy – *Funambulus palmarum* \(Indian palm squirrel\)](#)
- [NCBI Taxonomy – *Calotes versicolor* \(garden lizard\)](#)
- [NCBI Taxonomy – *Ixora coccinea*](#)
- [NCBI Taxonomy – *Hibiscus rosa-sinensis*](#)
- [NCBI Taxonomy – *Chlorophytum comosum*](#)
- [NCBI Taxonomy – *Plumeria rubra*](#)
- [Kew Plants of the World Online – *Nerium oleander*](#)
- [Kew Plants of the World Online – *Lagerstroemia speciosa*](#)